

Realization of the analogous Chua circuit and the qualitative research upon the systems approach to chaos

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Abstract

In attempt of verifying chaotic behavior in physical systems, non-linear electronic systems are considered. Having only a few interesting measurable parameters, Chua's electronic circuit allows any kind of chaotic analysis via the behavior of the solutions to the representative set of 3 ODEs as trajectories in phase-space. Trajectories could converge to an equilibrium point (stable or unstable), a limit cycle, or strange attractor in different way through phase-space. The interest of this paper, concerns the assembly of the Chua circuit and the changes in the qualitative behavior of the system, via a measurement and a simulation, as a specific parameter is varied, thus experimentally reinforcing several chaotic phenomena occurring in the physical nature.

I. Introduction

The term "chaos" was first used^[1] in order to describe certain dynamical systems and underline that even deterministic dynamics may show after sufficiently long times, irregular and unpredictable behavior, this fact may be traced back to the work of Henri Poincare, who investigated the long-unsolved solar-system stability problem^[2]. Hence the concept of chaos is informally defined as the gradual loss of predictability as the arrow of time moves forward.

In the early 60's Edward Lorenz provided an example of such a system in the form of a set of coupled differential equations (Appendix; [A.1]), describing the most minimalistic and ideal model of the atmosphere^[3]. The physical nature of Lorenz's system was questioned by his peers, stating that the model he had used was so crude and oversimplified that no real experimental confirmation could be made and that the phenomena could only be realized by abstract mathematical methods that hold no connection to the physical reality. A research group led by Takashi Matsumoto had taken on itself the mission of realizing an electrical circuit that would mimic Lorenz's equations^[4]. Several years later the circuit was completed and was due to be demonstrated to Leon Chua, a visiting professor at Matsumoto's lab. The demonstration failed due to a key component failure. This fact of nonsuccess led Chua to wonder if a much simpler circuit may be constructed, one that is not governed by Lorenz's equations but still gives rise to chaotic behavior.

Success finally came in 1983^[5] when Matsumoto (guided by Chua) observed a chaotic attractor¹ appearing to have interesting structures that are different from Lorenz's and Rössler's^[A.2] attractors (The only two ODE sets at that time which were widely accepted as being chaotic).

Chua's circuit (Fig. 1a) is a nonlinear electronic circuit, that traditionally contains four linear elements and one nonlinear element, namely, two capacitors, one inductor, one resistor and the nonlinear resistor called Chua's diode.

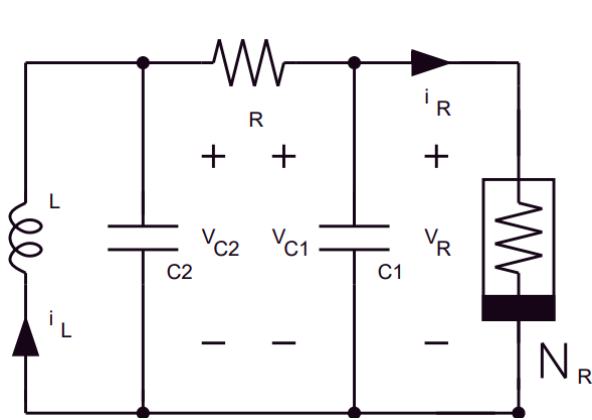


Figure 1a: Chua's circuit; L (-inductor), R (-resistor), $C_{1,2}$ (-capacitors), N_R (-Chua's diode - nonlinear component).

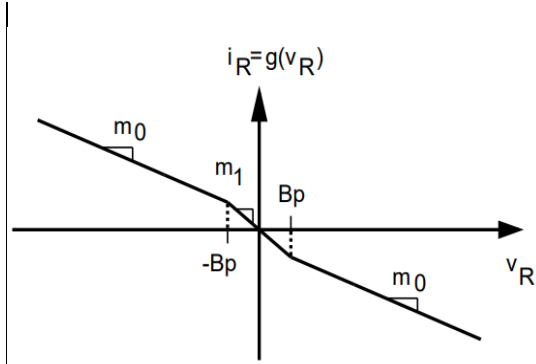


Figure 1b: The V-I characteristic of Chua's diode, which is divided into three piecewise-linear segments. The outer regions have slope m_0 ; the inner region has a slope m_1 with two breakpoints at $\pm B_p$.

The state equations for the circuit are:

$$(1) \quad \begin{cases} C_1 \frac{dv_{C_1}}{dt} = G(v_{C_2} - v_{C_1}) - g(v_{C_1}) \\ C_2 \frac{dv_{C_2}}{dt} = G(v_{C_1} - v_{C_2}) + i_L \\ L \frac{di_L}{dt} = -v_{C_2} \end{cases}$$

Where $G \equiv \frac{1}{R}$ and $g(\cdot)$ is a piecewise linear single-variable function given by:

$$(2) \quad g(v_R) = m_0 v_R + \frac{1}{2}(m_1 - m_0) \left[|v_R + B_p| - |v_R - B_p| \right]$$

¹An attractor is the value, or set of values, that a system settles towards over time; A chaotic system has a strange attractor, around which the system oscillates forever without repeating itself or settling into a steady state of behavior. It never produces the same value twice, and its structure is fractal, meaning the same patterns exist at every scale no matter how much we zoom into it. ^[6]

The relation above is shown in Fig.1b.

Chua's circuit as an autonomous circuit² made of standard components (resistors, capacitors, inductors), satisfying the three criteria of chaos^[7]:

1. one or more nonlinear elements,
2. one or more locally active resistors³,
3. three or more energy storage elements⁴,

makes it the simplest electronic circuit displaying chaotic phenomena.

Chua's circuit may be used to demonstrate many of the important features of non-linear dynamical systems such as period doubling bifurcations⁵, attractors and chaos in general.^[8]

II. Experimental

Realization of the analogous Chua circuit:

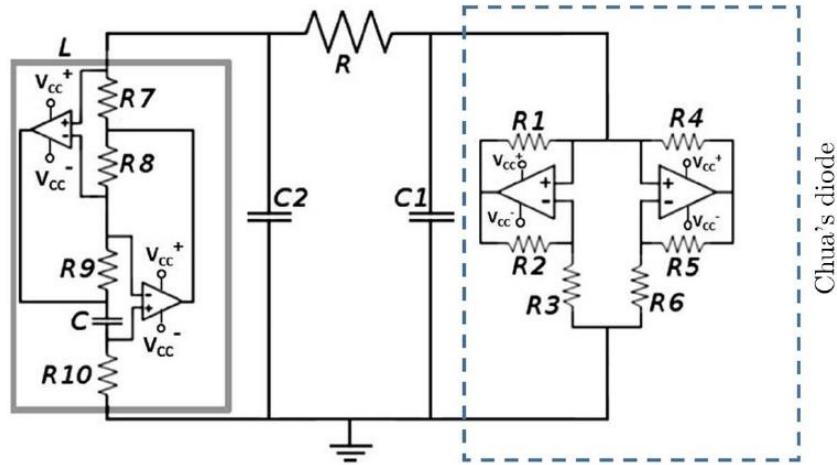


Figure 2: Chua's circuit schematic with realizable components

An analogous Chua circuit has been realized^[10] in which an inductor-gyrator is preferred over the inductor for reasons to be discussed later. The only non-trivial circuit element to implement is Chua's diode, that was synthesized by two op-amps and six resistors.^[10]

²An autonomous circuit is a circuit that produces a time varying output without a time varying input.

³A "locally active resistor" is a device that has negative resistance and is active (it can amplify), providing the power to generate the oscillating current. The locally active non-linear component was realized using Chua's diode.

⁴The energy storage elements are two capacitors and an inductor, as seen in Fig.1a.

⁵A period doubling bifurcation corresponds to the creation or destruction of a periodic orbit with double the period of the original orbit.

The implementation of Chua's diode and circuit is as seen in Fig.2. (full specification of parts and detailed schematics are shown in Appendix B). The configuration seen in Fig.2. yields Chua's set of equation (1) noticing that the inductance of the inductor-gyrator can be evaluated by:

$$(3) \quad L = \frac{R7 \times R9 \times C \times R10}{R8}$$

The fully constructed Chua circuit on which the experiments were performed may be regarded in Fig.3.

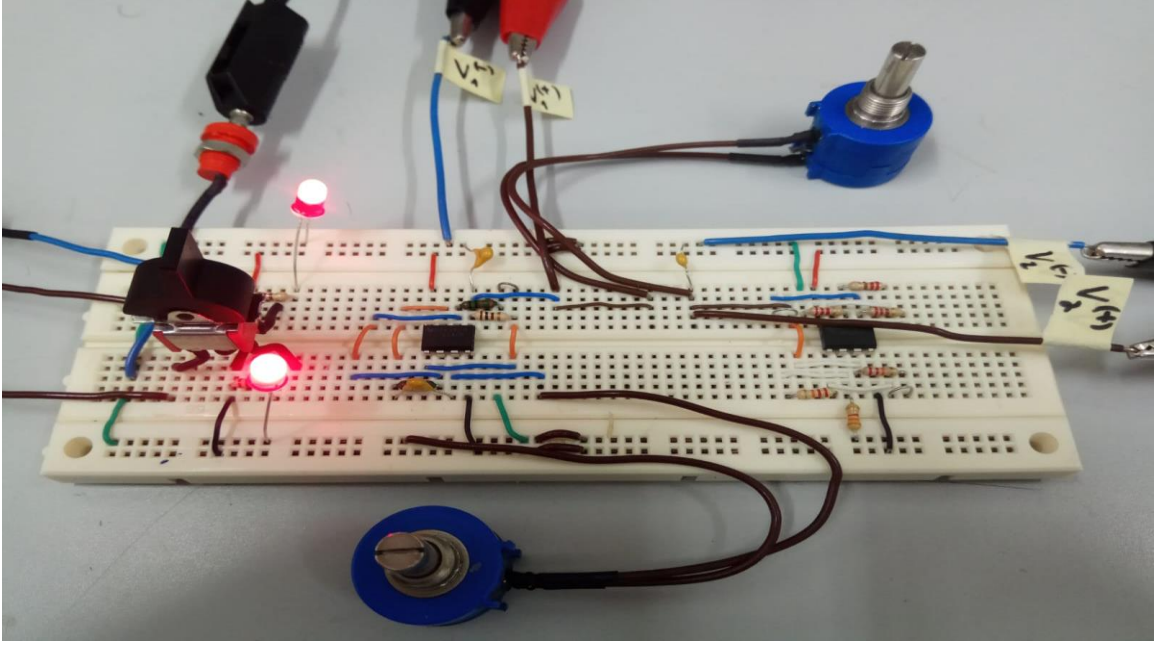


Figure 3: Fully constructed analogous Chua circuit; the inductor-gyrator and Chua's diode may be indicated by the left and right TL082 op amps regions respectively.

Experimental chaos from Chua's circuit

In this section, several chaotic phenomena and bifurcations occurring in Chua's circuit are shown in the systems approach towards chaos:

1. Period –doubling approach to chaos
2. intermittency approach to chaos
3. period adding bifurcations

In the following sections the control parameter under variation was the resistor R , ranging its resistance from $R = 2000\Omega$ to $R = 1500\Omega$, all other values were fixed and their values may be seen in [B.2]. The circuit was induced by 9V in both channels as power supply for the circuit.

Period-doubling approach to chaos

Upon switching the circuit on, a temporary trajectory was seen, that very quickly settles to a steady state response, this is the attractor of the system, meaning that the responding, or passive part of the system can take many different initial conditions all leading to the same steady state behavior. For any value of R in the range $(1500\Omega - 2000\Omega)$ the values of the current and voltage in the passive part return to the same values periodically with the same frequency of the driving force, thus observing a period-1 attractor or namely a limit cycle⁶. As the resistance is decreased a bifurcation is observed by means of period doubling.

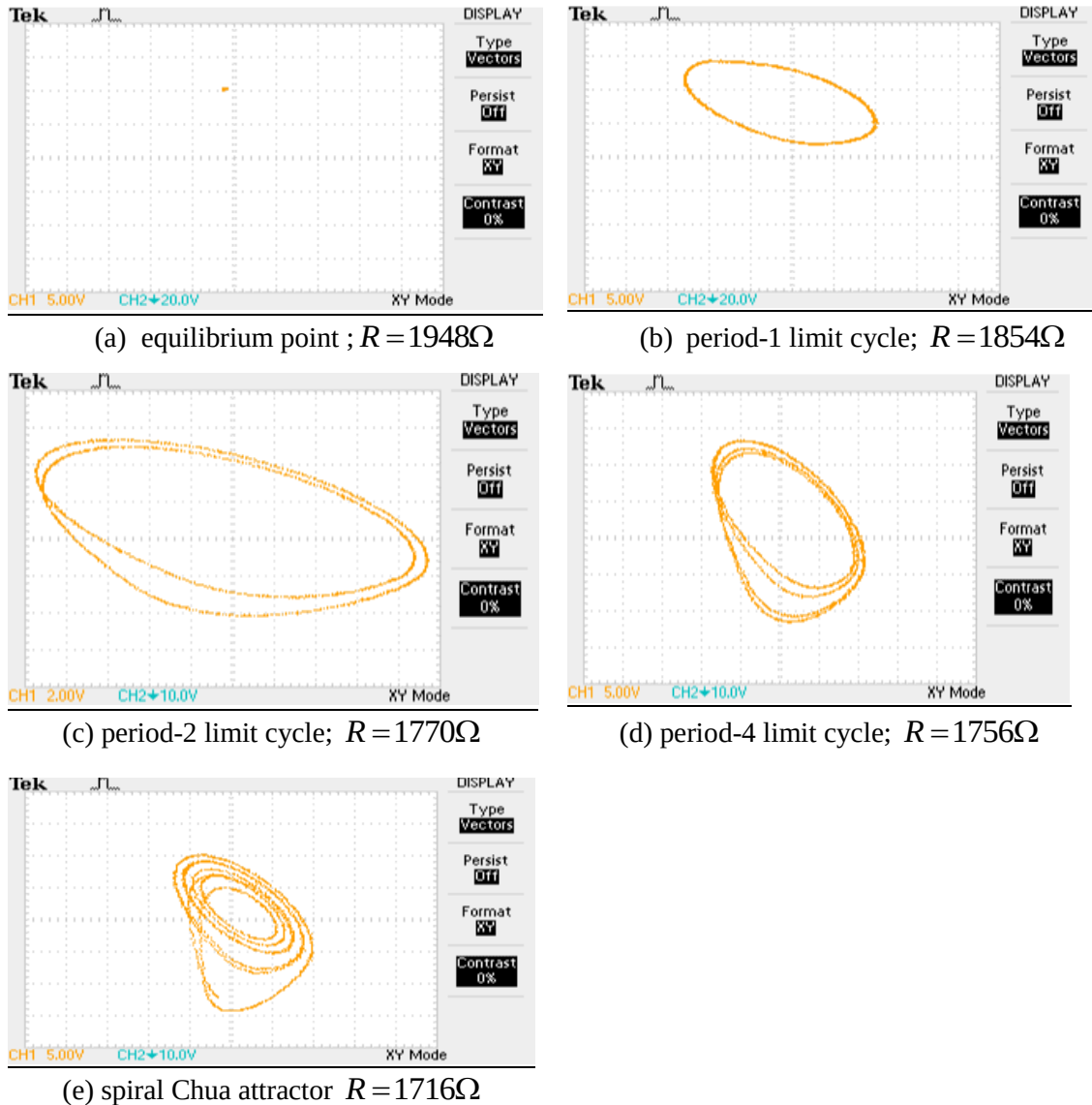


Figure 4(a)-(e): Phase portraits in the $V_{C1} - V_{C2}$ plane of period doubling sequences under the variation of R

⁶ A limit cycle is a closed trajectory in phase space, having the property that at least one other trajectory spirals into it either as time approaches infinity or as time approaches negative infinity. Limit cycles have been used to model the behavior of great many real world oscillatory systems

Intermittency approach to chaos

Smoothly decreasing R further beyond the first chaotic phenomena arising from period-doubling, a range of parameter values where a stable period-3 limit cycle is observed.

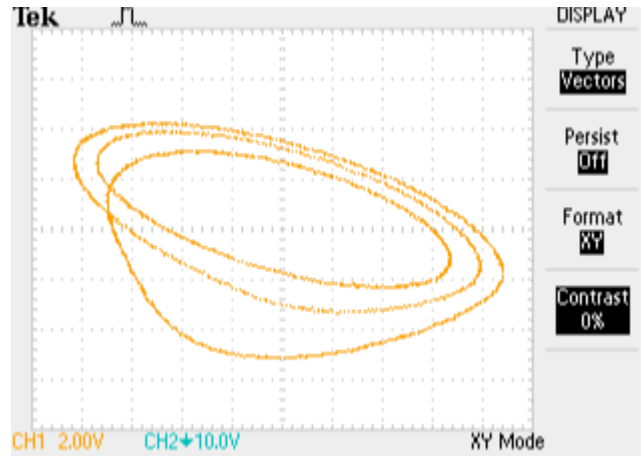
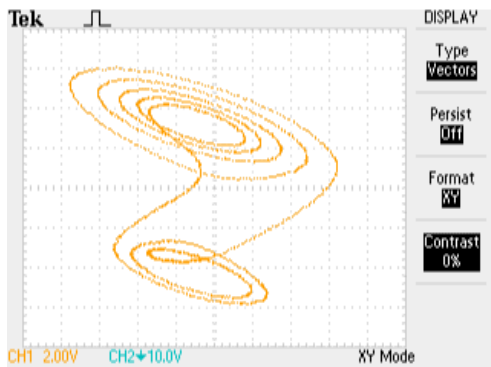


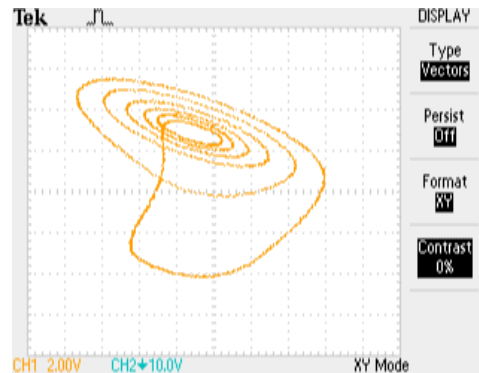
Figure 5: Intermittency near period-3 limit cycle, $R = 1693\Omega$

Period adding bifurcations

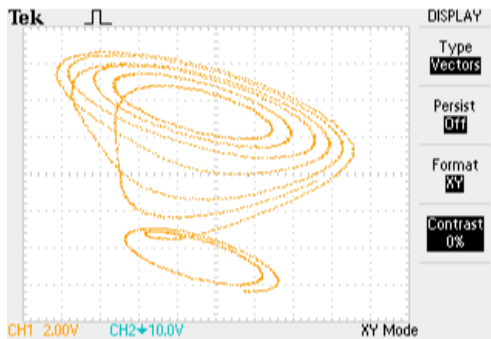
The phenomena of period adding is where windows of consecutive periods are separated by regions of chaos.



(a) Chua's double scroll; $R = 1648\Omega$



(b) 6-1 window; $R = 1595\Omega$



(c) Chua's double scroll $R = 1490\Omega$

Figure 5(a)-(c): Phase portraits in the $v_{C1} - v_{C2}$ showing the transition from a region of chaos through a stable 6-1 window into chaos again

III. Discussion

Realization of the analogous Chua circuit

The analogous Chua circuit was realized on a solderless breadboard (a point to be referred to later), where the inductor was replaced by an inductor-gyrator, the reasons taken into consideration for the elimination of the inductor are:

- The inductance value required for the realization of the circuit was not readily available.
- The parasitic⁷ resistance of the inductor may be destructive enough to prevent Chua's circuit from functioning as intended.
- Chua's circuit had not yet been implemented in its current inductor-less version, in this course.

Chua's diode was realized using the methodology in [11], thus providing the means to construct the diode using a minimal number of elements; using only two op-amps (integrated in one part) and six resistors (Fig.2).

For the duration of the experiment high sensitivity upon the control parameter and stable connections were crucial in obtaining proper results. The means of control was a potentiometer lacking the sensitivity needed, thus an attempt of constructing and applying a voltage divider was employed without great success, this fact affected all of our measurements as seen in the results, meaning not all phenomenon may be fully accounted for. Building the circuit on a breadboard resulting in unstable connections may be another reason taken into account for having an impact on the performance of the system and quality of results (during the experiments even the smallest of touches interfered).

The authors wish to state that future research of the system should probably be implemented by using a board that requires soldering for stable connections.

Experimental chaos from Chua's circuit

Period-doubling approach to chaos

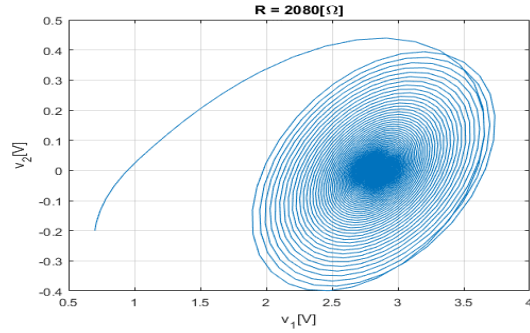
Period doubling is a phenomenon which is considered to be universal in non-linear dynamical systems, where the same qualitative behavior is observed in many physical systems.

In Chua's circuit route to chaos, as the resistance R is smoothly decreased an equilibrium point loses its stability and a stable limit cycle emerges, upon further decrease, the stable limit cycle eventually loses stability and a stable limit cycle of twice the period emerges. As R is decreased further the period-2 limit cycle loses its stability and a stable period-4

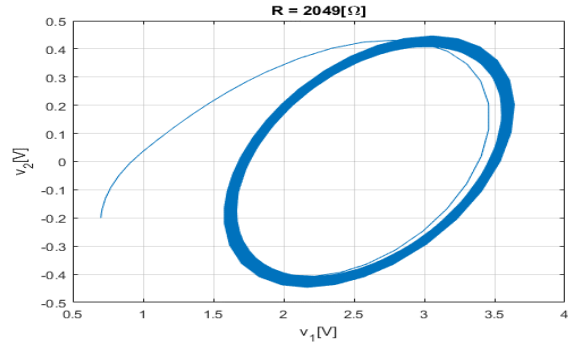
⁷ A parasitic element is a circuit element that is possessed by an electrical component but which is desirable for it to have for its intended purpose, thus an inductor may possess parasitic resistance.

limit cycle appears. This type of bifurcation converges to a limit, with the limit being a cycle of infinite period at which chaos is observed in form of a chaotic attractor. In practice (Fig.4(a)-(e)), the period-2 and period-4 were easily seen, a further small change in R yielded the single scroll, namely Chua's spiral attractor.

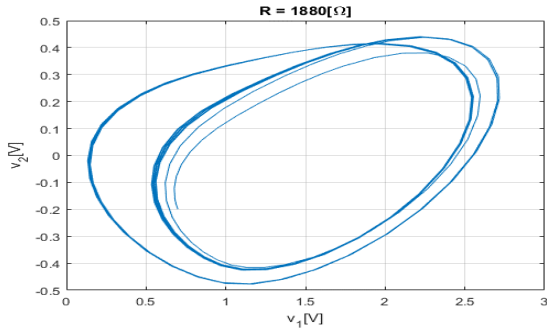
A simulation was built by numerically solving Chua's set of equations (1) with the same parameters as in the experiment, yielding a qualitative simulation of the period doubling route to chaos as seen in Fig.7.(a)-(e)



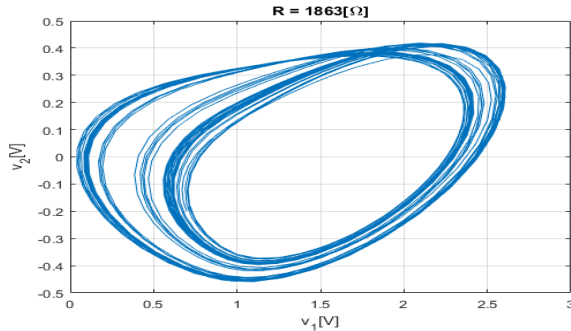
(a) equilibrium point ; $R = 2080\Omega$, 6.3% deviation from experimental value (Fig.4(a))



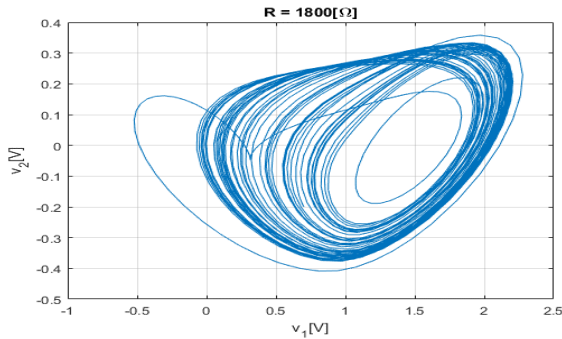
(b) period-1 limit cycle; $R = 2049\Omega$, 9.5% deviation from experimental value (Fig.4(b))



(c) period-2 limit cycle; $R = 1880\Omega$, 5.8% deviation from experimental value (Fig.4(c))



(d) period-4 limit cycle; $R = 1863\Omega$, 5.7% deviation from experimental value (Fig.4(d))



(e) spiral Chua attractor; $R = 1800\Omega$, 4.6% deviation from experimental value (Fig.4(e))

Figure 7(a)-(e): Simulation of the phase portraits in the $v_{C1} - v_{C2}$ plane of period doubling sequences under the variation of R ; each simulation (a)-(e) is accompanied with the deviation from the experimental values: (a)1948Ω (b)1854 Ω,(c)1770 Ω,(d) 1756 Ω,(e)1716 Ω; the initial condition of v_{C1}, v_{C2} where set to 0.7V and -0.2V respectively.

When running the simulation one can readily see that there are regions of stability where each limit cycle lives in, one can also observe the transitions from region to region (the phase portraits shown above where selected from within these regions). As seen above the experimental results (Fig.4(a)-(e)) and the simulations qualitatively show the same phenomena of period doubling bifurcations where the mean deviation is approximately 6%, thus capturing the chaotic essence of Chua's equations.

Intermittency approach to chaos

The key feature of this route is as follows ^{[8],[9]}; over a certain range of the control parameter, the dynamic system has a periodic orbit, as the control parameter is tuned beyond a critical value, some irregular short bursts appear among the long regular phases. As the value of the control parameter changes further the bursts appear more frequently and eventually the system moves into a chaotic regime.

Upon further decreasing R beyond Chua's spiral attractor, a stable period-3 was identified (Fig.5) accompanied with said bursts which drove the system into its chaotic regime, these bursts have not been captured experimentally due to the sensitivity of the control parameter. Thus the stable period was observed and (luckily) captured experimentally whereas the bursts have not. Employing the simulation built yielded no success, possibly due to the inability to measure in sufficient precision the initial conditions for the three phase space variables.

Period adding bifurcations

As the parameter R is varied a stable period- n orbit, $n=1, 2, \dots$ followed by a region of chaos, then a stable period followed again by chaos and so on. Two cases exist, the first is the case of Chua's attractor where period-3,4 are shown as seen in the period doubling rout (Fig.4(a)-(e)). The second case is the double scroll Chua's attractor region shown in Fig.6(a)-(c) in order of decreasing R . Because of the high sensitivity of the analogous system this region could not be sampled efficiently thus showing only one window, when in practice there were more.

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Appendix

A.1. Lorenz equations:

$$\dot{x} = -\alpha(x - y)$$

$$\dot{y} = \beta x - y - xz$$

$$\dot{z} = xy - \gamma z$$

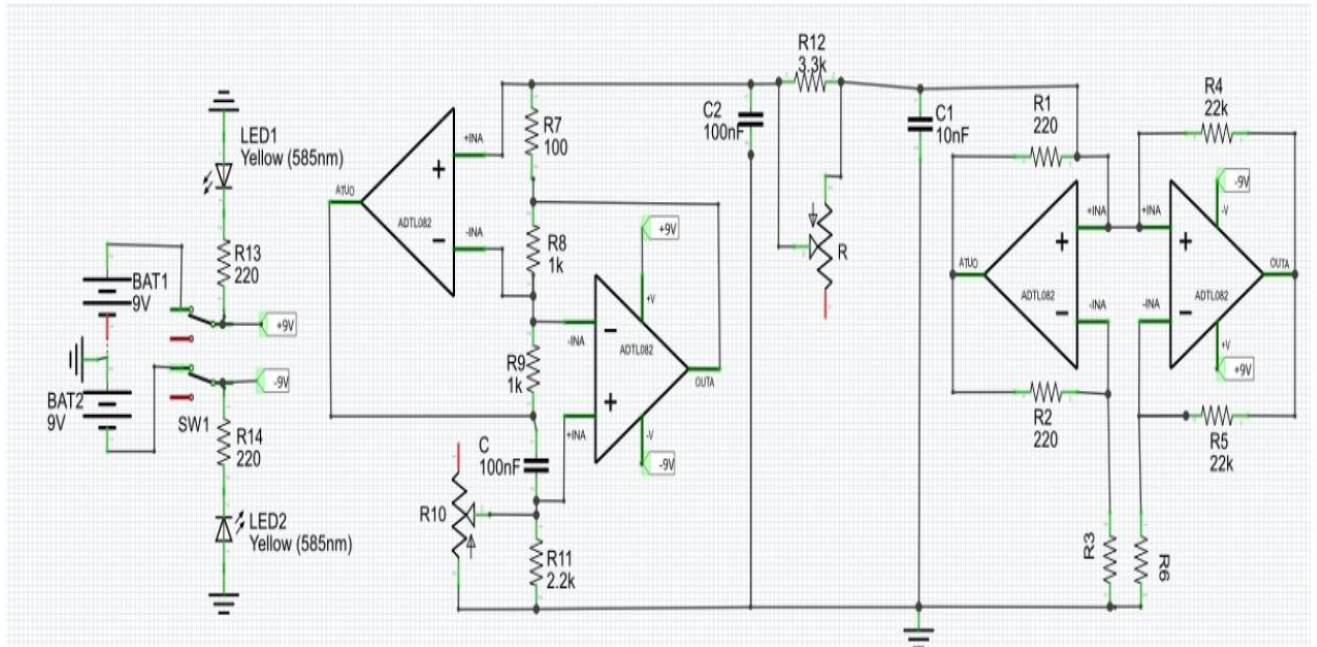
A.2. Rössler's equations

$$\dot{x} = -y - z$$

$$\dot{y} = x + \alpha y$$

$$\dot{z} = \beta + z(x - \gamma)$$

B.1. schematic of the analogous Chua circuit ^[9]



B.2. Specification of parts

$R1, R2$	$220\ \Omega$	$C1$	10nF
$R3$	$2200\ \Omega$	$C2$	100nF
$R4, R5$	$22000\ \Omega$	C	100nF
$R6$	$3300\ \Omega$	L	18mH
$R7$	$100\ \Omega$		
$R8, R9$	$1000\ \Omega$		
$R10$	$1800\ \Omega$		

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